

Detailed design of the ITER central solenoid

P. Libeyre*, N. Mitchell, D. Bessette, Y. Gribov, C. Jong, C. Lyraud

ITER Organization, Cadarache, St. Paul lez, 13108 Durance, France

ARTICLE INFO

Article history:

Available online 5 May 2009

Keywords:

ITER
Tokamak
Central solenoid
Coils
Hexapancake
Precompression structure
High voltage
Electrical insulation

ABSTRACT

The central solenoid (CS) of the ITER tokamak contributes to the inductive flux to drive the plasma, to the shaping of the field lines in the divertor region and to vertical stability control. It is made of 6 independent coils, using a Nb₃Sn cable-in-conduit superconducting conductor, held together by a vertical precompression structure. This design enables ITER to access a wide operating window of plasma parameters, up to 17 MA and covering inductive and non-inductive operation. Each coil is based on a stack of multiple pancake winding units to minimise joints. A glass–polyimide electrical insulation, impregnated with epoxy resin, is giving a high voltage operating capability, tested up to 29 kV. The CS performance is fatigue driven mainly by the stress levels in the conductor jacket and in the precompression structure needed to keep the modules in contact during the repulsive forces which can arise in operation. A rigid connection to the TF coils provided at one end and a centering support at the other end allow to resist net vertical forces as well as unbalanced radial forces while avoiding torsion transmission from the TF Coils to the CS assembly.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

The central solenoid (CS) is a part of the ITER magnet system [1]. It achieves three functions: production of the inductive flux to drive the plasma, shaping of the field lines in the divertor region and control of the vertical stability. These operating functions produce conflicting requirements on the coils. Optimisation of the requirements has led to the selection of 6 independent identical coils called modules, held together by a vertical precompression structure (Fig. 1) [2]. The present design satisfies the requirement of ITER, as an experimental device, to allow access to a wide operating window of plasma parameters, up to 17 MA and covering inductive and non-inductive operation. The CS main parameters are given in Table 1.

2. Plasma operation

At plasma initiation, all CS modules initially magnetized are discharged on resistors of the switching network units (SNU), inducing the toroidal electrical field inside the vacuum vessel, which produces breakdown of the gas, its ionization and generates plasma current. The reference scenario assumes that at this instant a total current of 21.91 MA is reached in each CS module with a maximum field on the conductor of 13 T and a maximum voltage of 10 kV induced by the discharge on resistors. To achieve plasma shaping

and current ramp-up, the current is then decreased differently in each CS module and even reversed in the central modules whereas it is still positive in the upper and lower modules, which results in repulsive forces between modules to be resisted by the precompression structure. The maximum current in the central modules is reached at the end of burn, but with a maximum magnetic field of 12.4 T.

3. Coil design

3.1. Conductor

The CS conductor (Fig. 2) is a cable-in-conduit conductor using Nb₃Sn strands and a JK2LB stainless steel jacket [3]. The conductor current (40–45 kA) is a compromise between structural issues (stress concentration aspect calls for decreasing the conductor size since Nb₃Sn is very sensitive to strain as shown with the model coils) and cable current density (thermal protection calls for larger current to lower the amount of copper required). As a structural material the conductor jacket has to resist the large electromagnetic forces arising during operation and to demonstrate a good fatigue behaviour. A manganese stainless steel, called JK2LB, was selected in order to withstand the planned 60,000 cycles (2 cycles per burn) during operation, taking advantage of its lower thermal contraction coefficient from room temperature to cryogenic temperature (0.21%) than that of 316LN (0.29%) to provide additional compression during cooldown. The conductor will be produced in unit lengths up to 910 m.

* Corresponding author. Tel.: +33 4 42 25 46 03; fax: +33 4 42 25 66 28.
E-mail address: paul.libeyre@iter.org (P. Libeyre).

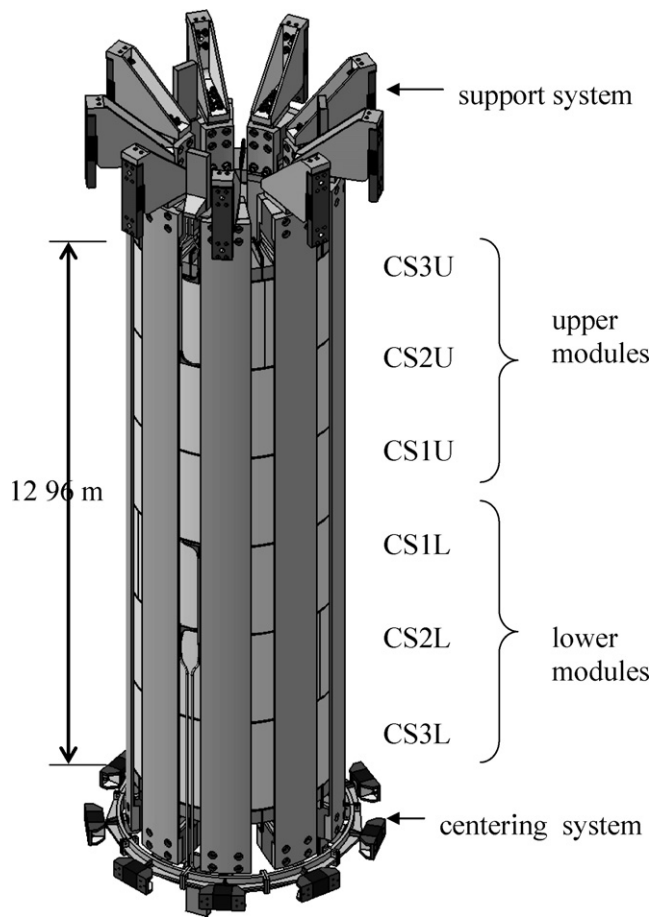


Fig. 1. The ITER central solenoid.

3.2. Winding-pack

The conductor is wound into multiple pancakes, each unit length of 910 m allowing the winding of a hexapancake. The winding-pack of one module is a stack of 6 hexapancakes with one quadpancake in the middle. Joints between adjacent multiple pancakes are located in the pancake transition area of the outermost turn of the pancake, the reference design being a butt joint fitting within restricted

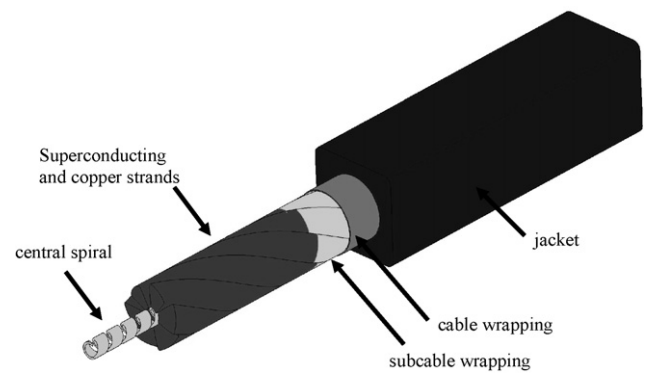


Fig. 2. The CS conductor.

space and avoiding any protrusion out of the coil outer diameter. This type of joint, used in the Central Solenoid Model Coil (CSMC), achieves very low resistance, in the range of 1.5 nΩ [4]. The design of the terminal area also relies on the use of a butt joint connecting the coil to the busbar extensions routed vertically along the outer diameter of the CS modules. Special attention was given in this area to the support of the outermost turn, which is locked to the last-but-one turn to withstand the outbursting hoop force arising when the coil is energized. Efficient cooling of the coil is provided by an internal circulation of supercritical helium flowing inside the conductor. Helium inlets are managed at the inner bore whereas outlets are managed at the outer diameter. The winding scheme is the same for each pancake except for the last pancake of the quadpancake, which has a longer filler at the outermost turn. Fillers are installed at the inner and outer diameters in the turn transition area and in the pancake transition area (Fig. 3). A glass-polyimide electrical insulation, wrapped around the turns and the modules after heat treatment of the hexapancakes and quadpancakes and impregnated with epoxy resin, is giving a high voltage operating capability (test voltage: 29 kV) [5].

3.3. Electrical supply

Each CS module is energized independently with its own power supply, with the exception of the central modules CS1U and CS1L which are powered in series with one single power supply. The busbar extensions of each module are connected to the current leads installed in the feeders located above and under the tokamak (Fig. 4). The three upper modules are connected to the upper

Table 1
ITER CS main parameters.

Number of modules	6
Module inner radius	1 342.0 mm (@ 293 K)
Module outer radius	2 095.5 mm (@ 293 K)
Module height	2 135.3 mm (@ 293 K)
Total height of assembled modules	12 961.8 mm (@ 293 K)
Number of turns per pancake	14
Module composition	6 hexapancakes + 1 quadpancake
Conductor type	Cable-in-conduit (square JK2LB jacket)
Superconducting material	Nb ₃ Sn
Conductor dimensions	49 mm × 49 mm (@ 293 K)
Conductor current	40–45 kA
Total current in one module	21.91 MA
Maximum induction on conductor	13.0 T (@ 40 kA in all modules)
Insulation system	Glass-polyimide + epoxy VPI
Coil test voltage	29 kV
Number of precompression structure units	9
Tie-plate and keyblock material	316LN stainless steel
Total weight (including precompression structure, support and centering system)	974.2 T

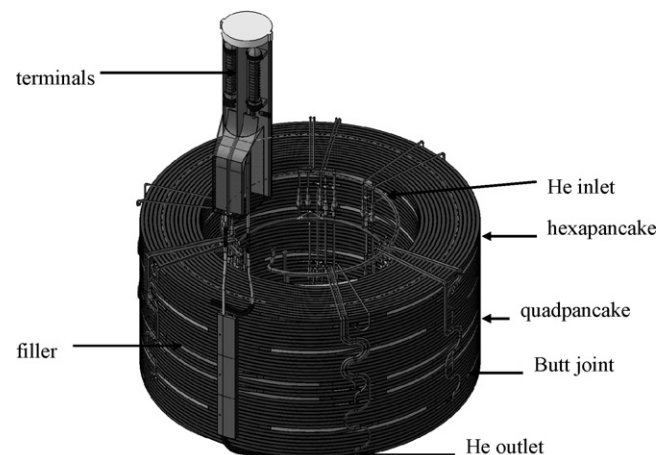


Fig. 3. CS module.

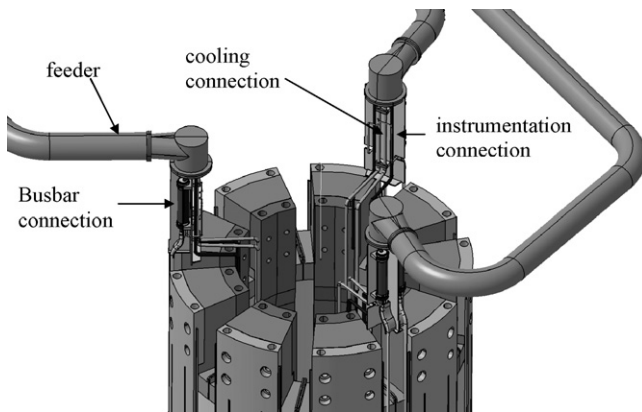


Fig. 4. Connection of the CS modules to the feeders.

feeders and the three lower modules are connected to the lower feeders.

3.4. Cooling

All modules are cooled in parallel. Manifolds are installed in the inner bore of the CS to distribute helium to the inlets and collect it from the outlets. Due to the tight space requirements at the outer diameter of the modules, the helium pipes are routed from the outlets first vertically towards the module upper or lower end and then routed in between adjacent modules towards the inner bore where they are connected to the main collector. Slots managed in the thick glass-epoxy plates inserted between modules allow pipes to cross radially the module without being squeezed during operation. Similarly to the busbar extensions, the main pipes of the three upper modules are connected to the upper feeders whereas those of the lower modules are connected to the lower feeders.

4. Precompression structure

4.1. Stack assembly

In the stack assembly the module terminals and busbar extensions are rotated from each other by 120° to allow connection of each module to its feeder terminal at top or bottom of the CS and the three lower modules are put in upside down position with respect to the upper ones so as to achieve symmetrical arrangement around the equatorial plane.

4.2. Precompression system

The precompression structure is needed to avoid separation of the modules during operation and must allow radial breathing of the stack assembly. It is designed as a set of nine clamping systems equally distributed around the stack assembly (Fig. 5). Each clamping system is made of two vertical tie-plates located at the inner and outer diameter of the modules and linked at top and bottom by two key blocks. Two buffer zones made of JK2LB are installed at top and bottom of the stack assembly to distribute evenly the pressure inside the modules. Precompression is applied at the end of the assembly using a jack and a temporary back plate fixed onto the upper side of the tie-plates. The jack applies pressure onto the upper key block and pulls upwards the tie-plates linked to the temporary back plate. Once the precompression is applied, wedges are adjusted and inserted radially in between the upper keyblock and the vertical tie-plates, allowing then removal of the jack and the back plate. An additional precompression is

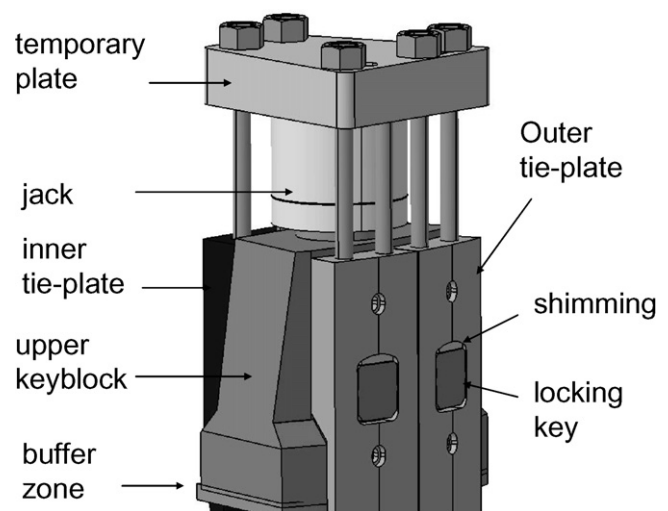


Fig. 5. Precompression structure.

provided during cooldown by the differential thermal contraction between the stainless steel tie-plates and the JK2LB buffer zones.

The mechanical analysis of the CS during operation shows that an average axial precompression of 26 MPa applied on the whole surface of the modules after cooldown prevents any vertical gap between modules from occurring. Most of this precompression (16.5 MPa) is achieved at room temperature, the limiting value being the maximum allowable stress in the tie-plates, an additional value of 9.5 MPa being provided by cooldown. The overall resulting axial extension of the tie-plates after cooldown is thus of 9 mm reducing to 3 mm at initial magnetization (IM), when the electromagnetic forces are compressing axially the stack assembly. The tensile stress in the tie-plates, initially of 138 MPa at room temperature, increases to 219 MPa after cooldown and decreases to a minimum of 45 MPa in the inner tie-plates and 53 MPa in the outer tie-plates at IM.

5. Support system

5.1. Loading conditions

During plasma operation the sum of the electromagnetic forces acting on the CS modules is not totally balanced due to the magnetic field created by the PF coils and the plasma itself, which results in a net vertical load directed either upwards or downwards, depending on the instant in the scenario. The maximum resulting load can thus vary from -31 MN up to $+54$ MN. Consequently the support system must be designed on one hand to withstand the weight of the CS and on the other hand to resist additional vertical loads in both directions.

5.2. Connection to the TF coils

The main principle of the CS support system is to hang the CS to the TF coils. Nevertheless, during operation the TF coils are experiencing a twisting deformation around the torus axis under the interaction of the TF current with the poloidal field. In order to avoid transmission of any torsional load from the TF coils to the CS, the support is thus located at one end of the TF coils and a centering system installed at the other end to keep the CS in vertical position. In the baseline design this support is provided at the top of the TF coils and the centering system located at the bottom whereas in an alternate design presently under study the support would be installed

at the bottom of the TF coils and the centering system at the top. The support system is split in nine units connected on one side to the precompression structure and on the other side to the upper or lower end of the TF coil case. In the reference design each unit is made of a corner bolted to a flexplate itself bolted to the upper flange of the coil case. In this design the flexplate is put in compression under downward loads. In the alternate design each unit is made of a link connected to the precompression structure through an axis and to the TF coil case with a square shaped horizontal key.

5.3. Centering system

The centering system is made of a set of nine units bolted to the TF coil case and connected to each other with an insulating plate in between. Each unit includes a spring loaded system allowing radial displacement and providing support against the net dynamic horizontal forces.

6. Conclusions

The detailed design of the central solenoid which has been developed combines modularity and tight arrangement in a restricted

narrow space to allow maximum flexibility during plasma operation. While modularity implied the use of a pancake winding solution, space restriction led to rely on butt joints located at the outer diameter and installation of helium distribution inside the bore of the modules. A modular precompression structure enables to achieve axial precompression while allowing radial expansion of the modules and prevents any separation of the CS modules under repelling forces arising during a scenario. Production of detailed drawings is underway.

References

- [1] N. Mitchell, P. Bauer, D. Bessette, A. Devred, R. Gallix, C. Jong, et al., Status of the ITER Magnets, *Fusion Eng. Des.* (2009), doi:10.1016/j.fusengdes.2009.01.006.
- [2] K. Yoshida, Y. Takahashi, N. Mitchell, D. Bessette, H. Kubo, M. Sugimoto, et al., Proposals for the final design of the ITER central solenoid, *IEEE Trans. Appl. Supercond.* 14 (June 2 2004) 1405–1409.
- [3] H. Nakajima, K. Hamada, K. Takano, K. Okuno, N. Fujitsuna, Development of low Carbon and Boron added 22 Mn–13 Cr–9 Ni–1 Mo–0.24 N steel (JK2LB) for jacket which undergoes Nb₃Sn heat treatment, *IEEE Trans. Appl. Supercond.* 14 (June 2 2004) 1145–1148.
- [4] Y. Takahashi, K. Yoshida, N. Mitchell, D. Bessette, Y. Nunoya, K. Matsui, et al., Performance of joints in the CS model coil and application to the full size ITER coils, *IEEE Trans. Appl. Supercond.* 14 (June 2 2004) 1410–1413.
- [5] P. Libeyre, B. Bareyt, I. Benfatto, D. Bessette, Y. Gribov, N. Mitchell, et al., Electrical design requirements on the ITER coils, *IEEE Trans. Appl. Supercond.* 18 (June 2 2008) 479–482.